



IPCop

Block Outgoing Traffic

Part—3

The first article in the series, which appeared in the December 2011 issue of *LFY*, discussed IPCop 1.4.21, its basic installation and configuration. Though an excellent firewall distro, basic IPCop has limited functionality. To enhance it, various add-ons are available. We covered the installation and configuration of four important add-ons in the January 2012 issue—'Advanced Proxy', 'URL Filter', 'Update Accelerator' and 'Calamaris'. In this third article in the series, we will cover the installation and basic configuration of 'Block Outgoing Traffic', one of the most important IPCop add-ons.

IPCop has an inherent lacuna in a very important area—it does not filter outgoing traffic based on TCP/UDP ports, but allows all outbound traffic (from the Green to the Red zone) on all ports. 'URL Filter' addresses the filtering of websites only. An interesting add-on for this functionality is 'Block Outgoing Traffic', also called BlockOutTraffic or simply BOT. With this, IPCop can filter outbound traffic based on Green IP/MAC addresses, Internet IP addresses and various services (ports), offering granular control over outgoing traffic, to conform with corporate Internet access policies. To achieve this, BOT creates Linux IPTables rules via the Web-based GUI. BOT enables IPCop to divide the Green network into groups based on their IP (or MAC) addresses. These groups are then configured to access only the desired services such as HTTP, FTP, telnet, SSH, DNS, etc, either singly or by forming service groups. Control can be further tightened by allowing the services or their groups only to required Internet addresses.

The parameters of a typical request from the internal network to the Internet include:

- Source: the IP address and the MAC address
- Destination: the IP address and MAC address (that of the default gateway)

- Source: Port number (TCP or UDP)
 - Destination: Services
- BOT allows the administrator to configure various groups by different parameters, as elaborated in Table 1.

Table 1: Groups

Group type	Group parameter
Internal computers (devices)	IP or MAC address
Destination	IP address
Services	Services required to access destination

(Source port number is considered not effective for filtering outgoing traffic, since it is generated by the requesting device.)

To establish the access policy, BOT creates IPTables rules to allow various groups of internal devices to access various groups of Internet destinations using groups of services.

Now let's configure a typical Internet access control scenario, to help understand BOT installation and configuration, which appears tricky only until you understand the underlying concept.

We assume the IPCop box was configured as in the previous articles, with TCP 445 for Web access, TCP 800 as proxy port, and the IP address 192.168.51.254.